Task 3: Temperature Monitoring System

1. Project Overview:

The purpose of this project is to design a simple IoT-based temperature monitoring system using Arduino. The system reads temperature values through a temperature sensor, displays the measured value on an LCD screen, and activates LED indicators to represent different temperature conditions.

2. Components Used

- Arduino Uno
- Breadboard
- Temperature Sensor (TMP36 / LM35 depending on what you used)
- LCD Display (16x2)
- 3 LEDs (Red, Yellow, Green)
- Resistors
- Jumper wires

3. Working Principle / System Logic:

The temperature sensor continuously measures the surrounding temperature and sends analog values to the Arduino board. The Arduino processes these values and displays the temperature on the LCD screen.

Three LEDs are used as indicators:

- Green LED turns on when the temperature is below 30 °C.
- Yellow LED turns on when the temperature is between 30 °C and 45 °C.
- Red LED turns on when the temperature is above 45 °C.

4. Screenshots:

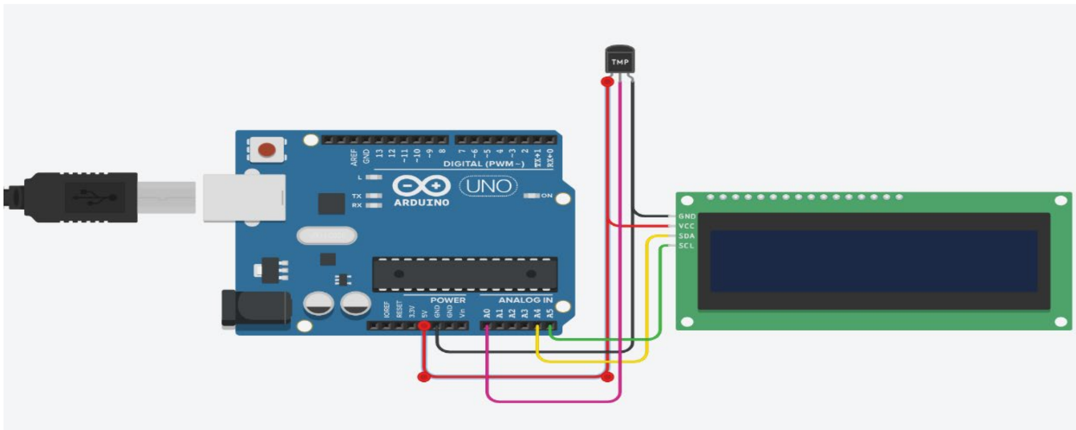


Figure1: shows the circuit design of the temperature monitoring system built using Arduino and connected components.

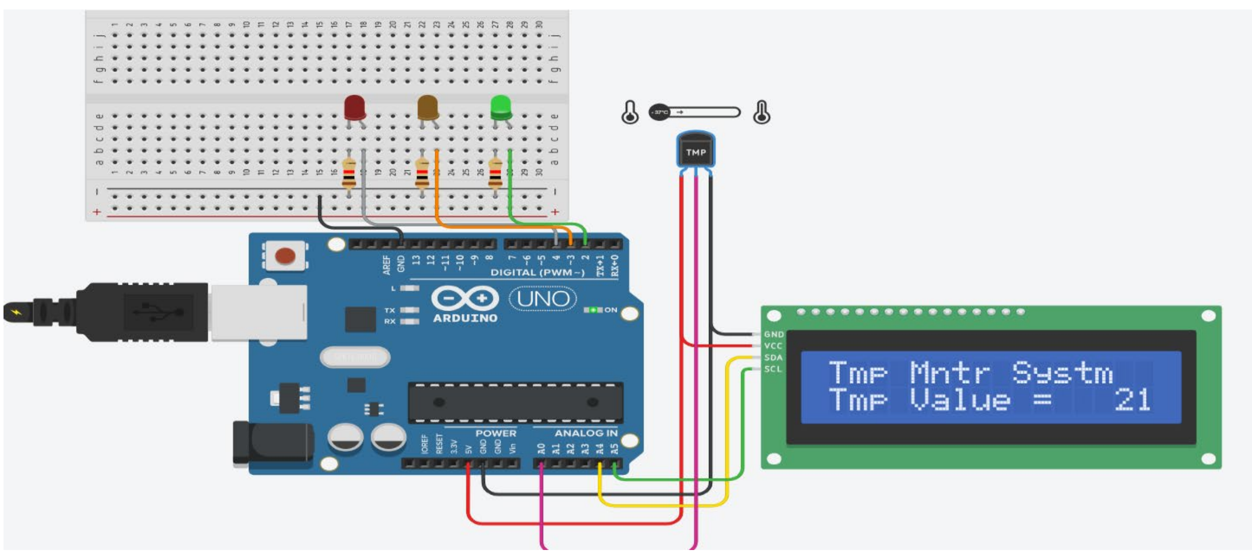


Figure 2: Temperature below 30°C→ Green LED ON

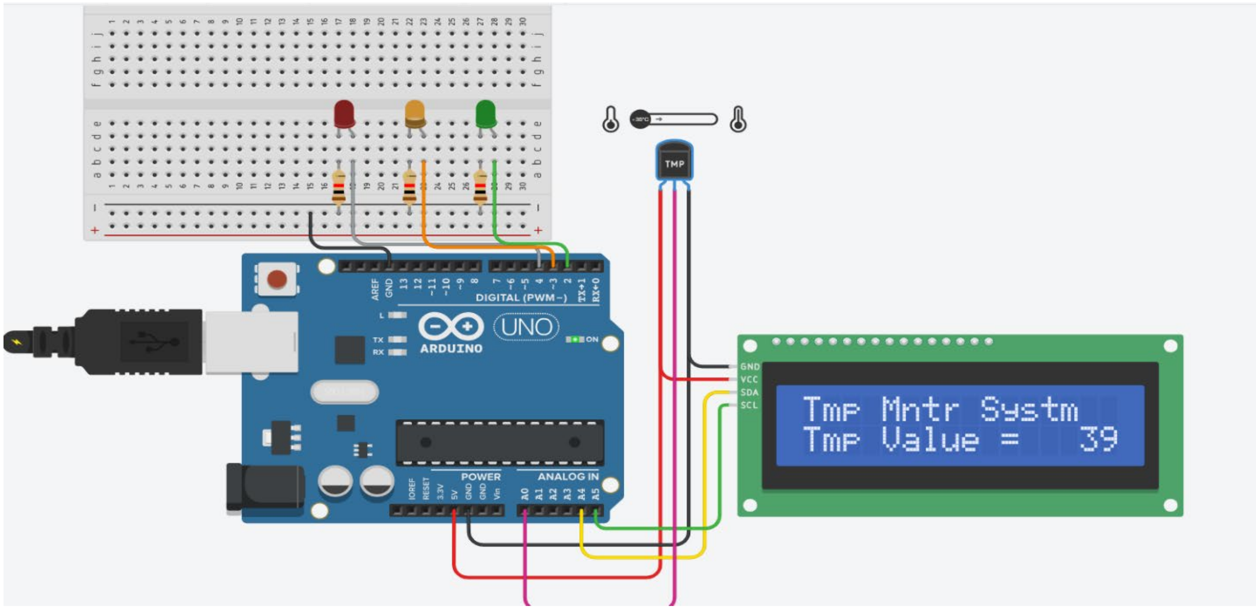


Figure 3: Temperature between 30°C and 45°C→ Yellow LED ON

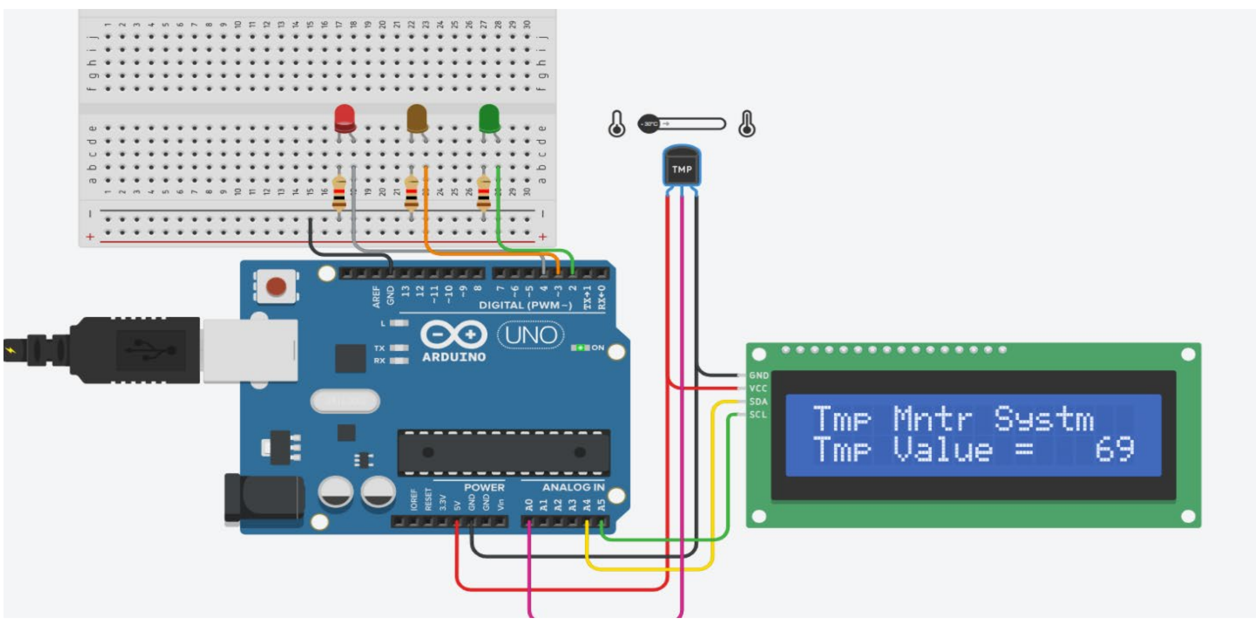
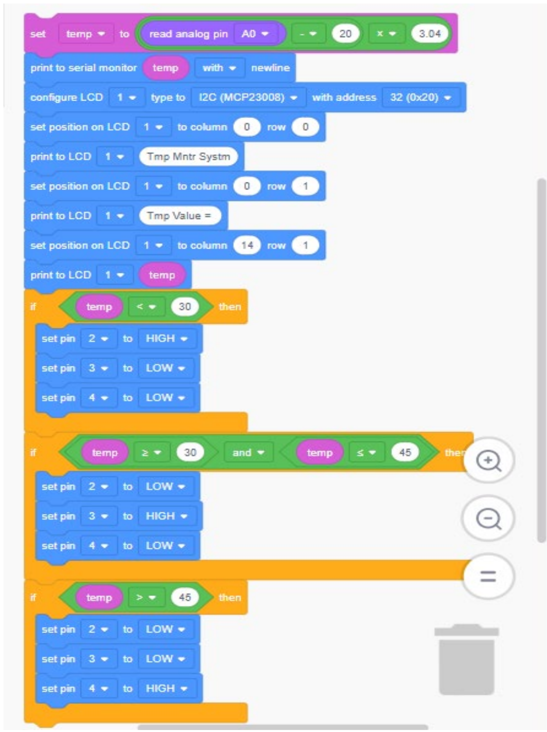


Figure 4: Temperature above 45°C→Red LED ON

5. Program Code

a) Arduino Blocks Code:



b) C++ Code:

```
#include <Adafruit_LiquidCrystal.h>
int temp = 0;
Adafruit_LiquidCrystal lcd_1(0);
void setup()
{
  pinMode(A0, INPUT);
  Serial.begin(9600);
  lcd_1.begin(16, 2);
  pinMode(2, OUTPUT);
  pinMode(3, OUTPUT);
  pinMode(4, OUTPUT);
}

void loop()
{
  temp = ((analogRead(A0) - 20) * 3.04);
```

```
  Serial.println(temp);
  lcd_1.setCursor(0, 0);
  lcd_1.print("Tmp Mntr Systm");
  lcd_1.setCursor(0, 1);
  lcd_1.print("Tmp Value = ");
  lcd_1.setCursor(14, 1);
  lcd_1.print(temp);
  if (temp < 30) {
   digitalWrite(2, HIGH);
   digitalWrite(3, LOW);
   digitalWrite(4, LOW);
  }
  if (temp >= 30 && temp <= 45) {
   digitalWrite(2, LOW);
   digitalWrite(3, HIGH);
   digitalWrite(4, LOW);
  }
  if (temp > 45) {
   digitalWrite(2, LOW);
   digitalWrite(3, LOW);
   digitalWrite(4, HIGH);
  }
  delay(10); // Delay a little bit to improve simulation performance
}
```

6. Conclusion

This project demonstrates a simple IoT prototype for temperature monitoring using Arduino. The system successfully measures temperature, displays data on and LCD screen, and provides visual alerts through LEDs. This type of project can be applied in smart homes, agriculture monitoring, and industrial environments.